

CASE 39

AUTOIMMUNE HEMOLYTIC ANEMIA

Autoimmune disease triggered by infection.

There are various ways in which an infection could induce autoimmunity: disruption of a tissue barrier might expose a normally sequestered autoantigen; the infecting microorganism might act as an adjuvant; microbial antigen might bind to self proteins and act as haptens; the microorganism might infect self-reactive lymphocytes (as in the case of infection by the Epstein-Barr virus) and induce autoantibody production; and the microorganism might share cross-reactive antigens with the host (molecular mimicry). The major histocompatibility antigens carried by an individual may also confer susceptibility to, or protection against, certain autoimmune diseases that can be incited by infection (Fig. 39.1).

There are only a few conditions in which an infection has been identified as the direct cause of the autoimmune disease. The longest established is rheumatic fever, which in the pre-antibiotic age was a frequent long-term consequence of infection with *Streptococcus pyogenes* (scarlet fever). Some strains of *S. pyogenes* lead to a type III autoimmune disease (caused by immune-complex formation; see also Case 36) of the renal glomeruli called post-streptococcal acute glomerulonephritis. These streptococcal strains are said to be nephrogenic (causing nephritis). About 3-4% of children infected with a nephrogenic strain of streptococcus will develop acute glomerulonephritis within a week or two of the onset of the streptococcal infection. What predisposes this subset of children to develop the complication is unknown.

TOPICS BEARING ON THIS CASE:

Humoral autoimmunity

Mechanisms for breaking tolerance

Coombs test

Fig. 39.1 Association of infection with autoimmune diseases. Several autoimmune diseases occur after specific infections and are presumably triggered by the infection. The case of post-streptococcal disease is best known. Most of these post-infection autoimmune diseases also show susceptibility linked to the MHC.

Associations of infection with immune-mediated tissue damage		
Infection	HLA association	Consequence
Group A <i>Streptococcus</i>	?	Rheumatic fever (carditis, polyarthritis)
<i>Chlamydia trachomatis</i>	HLA-B27	Reiter's syndrome (arthritis)
<i>Shigella flexneri</i> , <i>Salmonella typhimurium</i> , <i>Salmonella enteritidis</i> , <i>Yersinia enterocolitica</i> , <i>Campylobacter jejuni</i>	HLA-B27	Reactive arthritis
<i>Borrelia burgdorferi</i>	HLA-DR2, -DR4	Chronic arthritis in Lyme disease

A direct association of infection with autoimmune disease also occurs in patients with pneumonia caused by *Mycoplasma pneumoniae*, the case we shall discuss here. About 30% of patients with this infection develop a transient increase in serum antibody to a red blood cell antigen, and a small proportion of these patients develop hemolytic anemia, a type II autoimmune disease. Other type II autoimmune diseases are listed in Fig. 40.1. In this case, the decrease in the number of red blood cells (anemia) results from their immunological destruction (hemolysis) as a result of the binding of an IgM autoantibody to a carbohydrate antigen on the red cell surface. When the infection subsides, either spontaneously or after treatment, the autoimmune disorder disappears.

The IgM autoantibody agglutinates red blood cells at temperatures below 37°C. The antibodies are therefore called cold hemagglutinins or cold agglutinins, and the hemolytic disorder is known as cold agglutinin disease. Low titers of cold agglutinins (detectable at up to 1:30 dilution of serum) occur in healthy people. In about one-third of patients with mycoplasma infection there is a transient increase in the titer, but without symptoms; the rise usually goes unnoticed unless the patient happens to have a blood test, when clumping of the erythrocytes will be apparent.

The autoantibodies in the transient cold agglutinin syndrome resemble those in the chronic autoimmune disorder known as chronic cold agglutinin disease, which is of unknown etiology. This runs a protracted course in which the autoantibodies are characteristically oligoclonal, and lymphoproliferative disorders commonly develop. Thus chronic cold agglutinin disease behaves as a variant of an IgM gammopathy called Waldenström's macroglobulinemia. A gammopathy is an abnormal monoclonal or oligoclonal spike in the immunoglobulin electrophoretic pattern.

The case of Gwendolen Fairfax: the sudden onset of fever, cough, and anemia.

Gwendolen Fairfax was a healthy, unmarried 34-year-old who worked as a manager in a bank. She had never had any illness other than minor colds and the usual childhood infections. After developing a feverish cough her symptoms got progressively worse over the next few days and she decided to seek the advice of her physician, Dr Wilde.

Dr Wilde noted that Gwendolen was extremely pale; this was particularly noticeable in the palms of her hands, which were completely white. (Look at your palms; they have a healthy reddish pink color, unless your hemoglobin level is less than

34-year-old female with
respiratory infection; anemia.

10 g dl⁻¹). Her temperature was raised, at 38.5°C, and her respiratory rate was 30 per minute (normal 20). On listening to her chest, the physician heard scattered rhonchi (harsh breath sounds) at the bases of both lungs.

The physician ordered blood counts to be performed immediately. Gwendolen's hematocrit and hemoglobin level were low—hematocrit 26% (normal 38–46%) and hemoglobin 9.5 g dl⁻¹ (normal 13–15 g dl⁻¹). Her white blood cell count was elevated at 11,300 μl⁻¹, with 70% neutrophils, 24% lymphocytes, 6% monocytes, and 1% eosinophils, revealing an increased absolute neutrophil count, and the platelet count was 180,000 μl⁻¹. The chest radiograph showed patchy infiltrates in both lower lungs (Fig. 39.2). Dr Wilde suspected a diagnosis of mycoplasma pneumonia (also known as atypical pneumonia because there was no complete opacity over a whole lobe of the lung such as typically occurs in pneumococcal and other bacterial pneumonias). He advised Gwendolen to go to Reading Hospital.

In the hospital, a sputum sample proved negative for bacteria but contained abundant neutrophils. A blood sample was tested for the presence of antibodies against the red blood cells. In the presence of sodium citrate as an anticoagulant, the sample was chilled to 4°C; the red cells became clumped and the blood looked as though it had clotted. When warmed to 37°C, however, the red cells dispersed. Thus, the red cells were reversibly agglutinated in the cold. A blood sample was spun in a warmed centrifuge, and the red cells and plasma were separated. Gwendolen's red cells were tested using direct and indirect Coombs tests for the presence of anti-red cell IgG antibody (see Fig. 44.2); these tests proved negative. Her red cells, however, were strongly agglutinated by an antibody against the complement component C3.

When Gwendolen's plasma, or dilutions of the plasma, were incubated in the refrigerator with red cells from type O normal blood, the red cells were strongly agglutinated; when the same test was performed with red cells obtained from an umbilical cord in the delivery room of the hospital, agglutination was weaker, or negligible.

Gwendolen was started on erythromycin. In 3 days her symptoms had abated; she had no fever and her cough had greatly improved. Her hemoglobin had risen to 12 g dl⁻¹. She was discharged from the hospital, and advised to continue erythromycin orally and to avoid exposure to cold weather. When she returned to Dr Wilde's office 2 weeks later for a check-up, her fever had not returned, her cough had gone, and her hemoglobin was now a normal 14.5 g dl⁻¹.



Fig. 39.2 Chest radiograph of a patient with mycoplasma pneumonia. Patchy opacities (arrowed) are evident in the lungs. Courtesy of T. Griscom.

Pneumonia with marked anemia; hemolytic anemia following mycoplasma infection?

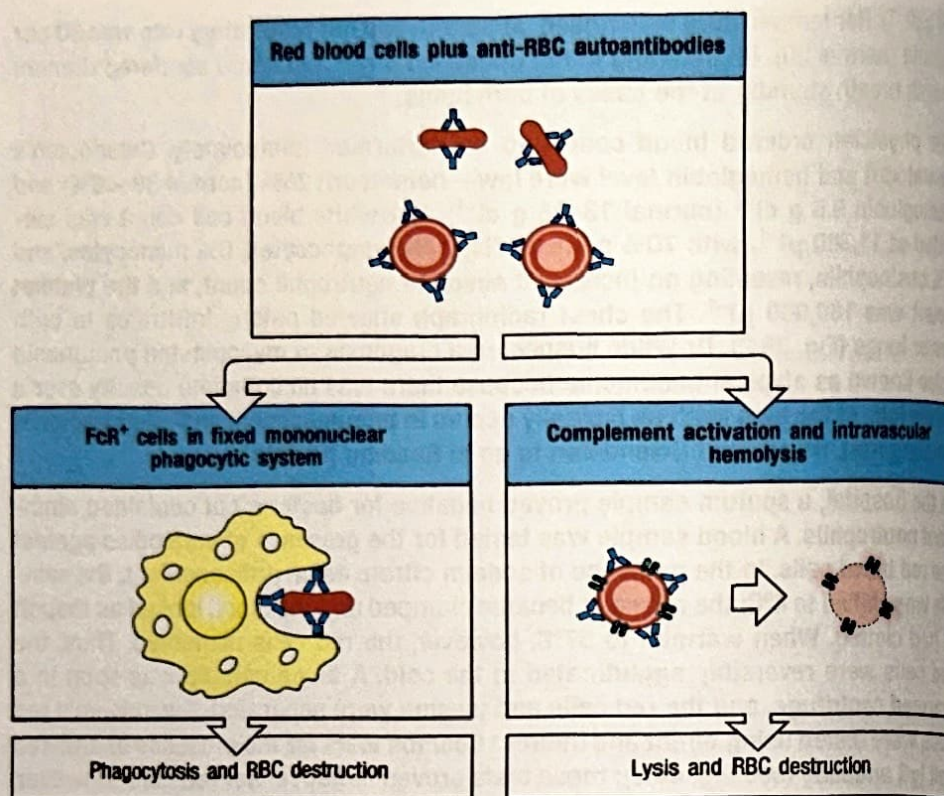
Autoimmune hemolytic anemia.

Gwendolen's autoimmune hemolytic anemia was transient, persisting only as long as the infection persisted. In contrast, autoantibodies against red blood cells in chronic autoimmune disorders of unknown origin tend to be persistent. Transient autoantibodies most frequently develop after nonbacterial infections, such as infectious mononucleosis (see Case 43). Autoimmune hemolytic anemia can also be induced by certain chemicals and drugs.

The autoantibodies in autoimmune hemolytic anemias are of the IgM or IgG class. They bring about hemolysis of the red blood cells either by complement fixation or adherence of the red cells to the Fcγ receptors on cells of the fixed mononuclear phagocytic system—principally in the spleen, but also in the liver and other organs (Fig. 39.3).

Anti-erythrocyte antibodies can differ in their physical properties. Some bind to the red blood cells at 37°C (warm autoantibodies); some bind at lower temperatures (cold hemagglutinins or hemolysins, like those in Gwendolen's case). In 1904, Donath and Landsteiner (who later received the Nobel Prize for the discovery of the ABO blood groups) reported antibodies in the blood of patients with syphilis that

Fig. 39.3 Autoantibodies specific for the surface antigens of red blood cells can cause hemolytic anemia. Antibody-coated red blood cells (RBCs) can be destroyed in several ways. Cells coated with IgG autoantibodies are cleared predominantly by uptake by Fc receptor-bearing macrophages in the fixed mononuclear phagocytic system (left panel). Cells coated with IgM autoantibodies and complement are cleared by macrophages bearing complement receptors CR1 and CR3 in the fixed mononuclear phagocytic system (not shown). The binding of certain rare autoantibodies that fix complement extremely efficiently causes the formation of the membrane-attack complex on the red cells, leading to intravascular hemolysis (right panel).



bound to human red blood cells at cold temperatures. These antibodies differ from cold agglutinins in that they cause the lysis of erythrocytes by complement when the blood is warmed to 37°C. These antibodies are still called Donath-Landsteiner or DL antibodies.

A transient increase in cold agglutinins is most frequently encountered during mycoplasma infection, infectious mononucleosis (the result of infection with the Epstein-Barr virus; see Case 43), and, rarely, during other viral infections. In mycoplasma pneumoniae the cold autoantibodies react with a red cell surface antigen called the I antigen. This is a branched carbohydrate structure that occurs on glycoproteins and glycolipids (Fig. 39.4). Cord blood cells express only small amounts of the I antigen and that is why Gwendolen's plasma did not agglutinate them, or did so only weakly. Cord erythrocytes express a related antigen, I (see Fig. 39.4); the I antigen is not fully expressed on red blood cells until 6 months of age.

It has not been possible to find with certainty a cross-reacting I antigen on *Mycoplasma pneumoniae*, which would provide a simple explanation for this autoimmune disease. Exactly how *M. pneumoniae* triggers the transient production of an autoantibody directed to red cells is not known, although there is a strong biochemical similarity between the attachment site (ligand) for the mycoplasma on the host cell surface and the red cell antigen to which the autoantibody binds (Fig. 39.5).

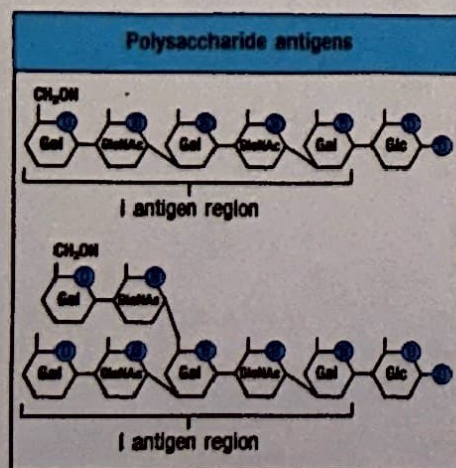


Fig. 39.4 Structure of the I and I antigen regions with which cold agglutinins react. Gal, galactose; GlcNAc, N-acetylglucosamine.

Questions.

1 The hemolytic anemia in Gwendolen Fairfax's case was not severe and it subsided rapidly. However, in other similar cases the hemolysis can be extensive and the drastic fall in hemoglobin can be life threatening. What emergency measure might be undertaken to stop severe hemolysis?

2 Autoimmune hemolytic anemia is caused, in general, by autoantibodies of the IgG and IgM classes. Could antibodies of other classes such as IgA, IgD, and IgE cause autoimmune hemolytic anemia?

3 In the direct and indirect Coombs tests (see Case 44), agglutination of the antibody-coated red blood cells is brought about by an antibody against human IgG that cross-links the IgG on the red cell surface and causes the cells to agglutinate. In the so-called 'non-gamma' Coombs test an antibody against C3 or C4 is used to agglutinate red blood cells coated with an IgM antibody that has fixed complement. Why is it important to have an antibody against the C3d or C4d portions of C3 or C4 to carry out the test?

4 Most normal individuals have a low titer (up to 1:30) of IgM anti-I cold agglutinins in their blood. As mentioned above, the I antigen is little expressed until 6 months of age. We must develop peripheral B-cell tolerance to this antigen in infancy. How does this happen? How might the mycoplasma infection break this tolerance?

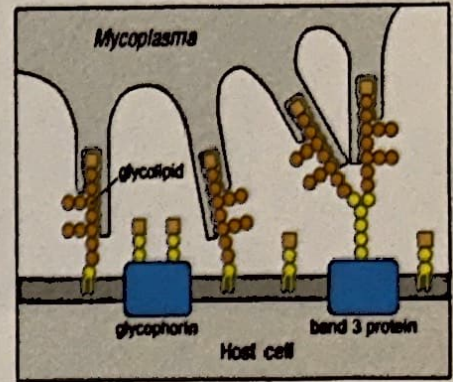


Fig. 39.5 *Mycoplasma pneumoniae* interacts with long carbohydrate chains of I antigen type on the human red cell membrane. These chains include the carbohydrates of band 3 protein and certain glycolipids, but not glycolipids and glycoproteins with short carbohydrate chains. Sequences of sugars that act as I antigen are shown as orange circles; the orange squares depict sialic acid.